Road Accident Severity Prediction

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March 2026

1 Introduction

Road accidents are a frequent and serious public safety concern worldwide, leading to significant human, social, and economic costs. Understanding the factors that contribute to the severity of accidents can help authorities implement more effective safety measures and reduce fatalities and injuries. In this project, we aim to apply machine learning techniques to analyze how various factors, such as environmental conditions, characteristics of the accident victims, vehicle features, and the type of accident, can influence and predict the severity of the outcomes for those involved.

2 General Organisation of the Work

In the ML project folder, you can find this report covering the entire work, the `previous_work_kaggle` folder, which contains the notebooks and associated dataset for the previous work part (detailed later in Section 3.2.3, *Reproducing Results*), and the most important folder, `Road_Accident_Severity_Prediction`, which contains all our work.

In the folder `Road_Accident_Severity_Prediction`, you can find the following components:

- The `data` folder contains the raw data in its original format, organized by year from 2019 to 2024. For each year, there are four tables with different characteristics of road accidents for that year.
- Four Python notebooks: `usagers.ipynb`, `lieux.ipynb`, `caract.ipynb`, and `vehicules.ipynb`. Each notebook processes its respective set of Excel files independently of the others. In `lieux.ipynb`, `caract.ipynb`, and `vehicules.ipynb`, we join the respective entries with the gravity label from the `usagers` table to include the target value. This structure helps to keeps the notebooks manageable in size and divided by parts.
- The notebook `all_features.ipynb` joins all tables together to produce the final unified dataset.
- The `data_clean` folder stores each table (now grouped across all years) after processing by the notebooks, separated by table type, and final joined cleaned dataset.
- Finally, `main.py` runs the full pipeline on the cleaned dataset, including hyperparameter tuning and final model evaluation. The `models` folder stores tuning results, while `results` contains the test-set evaluation of the best models.

3 Dataset

3.1 Dataset Selection and Source

After searching through government websites, we found a very useful database on data.gouv.fr: "Annual databases of bodily injury road traffic accidents – Years 2005 to 2024" (in french/original: "Bases de données annuelles des accidents corporels de la circulation routière – Années 2005 à 2024"). The data is collected by law enforcement units (police, gendarmerie, etc.) who intervene at the accident scene and record information about the accident.

For our analysis, we focus on the years 2019 to 2024. We chose this period because the format of the data changed slightly in 2019, making it more consistent, and we also wanted to work with recent data. Overall, the dataset already contains more than 700,000 entries, which provides a rich dataset for machine learning.

For each year, the database is divided into four separate CSV files, each covering a different type of information:

- Characteristics (CARACTÉRISTIQUES) – describes the general circumstances of the accident.
- Places (LIEUX) – describes the main location of the accident, even if it happened at an intersection.
- Vehicles (VÉHICULES) – contains information about the vehicles involved.
- People (USAGERS) – contains information about the people involved in the accident.

This is raw data, so it may contain errors or inconsistencies that we are going to handle further in the project.

3.2 Bibliography and state of the art

3.2.1 Dataset context

Our dataset contains approximately 700,000 records of road accidents in France. Most columns are categorical and some of them are numerical.

This dataset is commonly used for predicting accident severity and for forecasting the likelihood of an accident occurring at a given location, for ranking automobile safety by car brand in France, and for visualizing trends in road mortality based on the type of road user (pedestrians, cyclists, motorcyclists, drivers), as well as age and gender. It has also been used in studies analyzing the most dangerous intersections, the most hazardous times of day, and the effectiveness of car safety equipment.

Typical preprocessing steps include handling missing data, removing outliers in numerical values, encoding categorical features, and performing feature selection and engineering to create more usable variables for analysis.

3.2.2 Usual ML techniques

For our binary classification task, common methods include logistic regression, decision trees, random forests, gradient boosting, Linear Discriminant Analysis (LDA), Quadratic Discriminant Analysis (QDA), and neural networks.

However, as it will be shown later, our dataset contains a large number of categorical variables, which after one-hot encoding become a high-dimensional and sparse matrix. Thus we have decided to reduce our set of ML techniques and focus on models that are more adapted to this type of data.

More precisely, logistic regression, which is commonly used for binary classification and is still effective with sparse encoded variables. Tree-based methods, namely decision trees for initial small experiments, and random forests and gradient boosting for larger tests, as they capable to capture non-linear dependence.

But methods such as LDA and QDA were excluded. In fact, they rely on stronger distributional assumptions and covariance structure estimation, and are less appropriate in a high-dimensional sparse case. Neural networks were also not considered, as they need heavier tuning, much more computational power to train, and are harder to interpret.

3.2.3 Reproducing results

We reviewed similar works on Kaggle where participants attempted to predict accident severity for different years.

Our approach differs in several ways. Our goal is to perform a more accurate and comprehensive analysis. We study how adding more information progressively improves the prediction of accident severity. We apply a broader set of machine learning methods with tuning and evaluate performance using multiple metrics. Certain features will be encoded differently to better represent them; for example, we will transform the hour of the day and the day of the year into continuous sine and cosine values to take into consideration their cyclical nature.

For our dataset (covering recent years), we found only one exploratory analysis that visualizes the distribution of different features, but it does not include a predictive modeling task. Most previous studies on accident severity prediction focus on the period from 2005 to 2016.

Since no predictive models have been applied to our time period, we refer to results from 2005–2016 as an indicative benchmark. However, this comparison is not strictly fair, as the distribution and nature of accidents may have changed over time.

In particular, the proportion of severe cases is slightly higher in the earlier dataset compared to ours.

Dataset	**Non-severe**	**Severe**
2005 - 2016 (3,499,072 samples)	0.792	0.208
2019 - 2024 (621 399 samples)	0.814	0.186

Table 1: Comparison of class proportions (binary classification)

We also compare the distribution across the four original severity categories:

Dataset	**Unharmed**	**Minor injury**	**Hospitalized**	**Fatal**
2005 - 2016	0.445	0.347	0.185	0.023
2019 - 2024	0.409	0.405	0.158	0.028

Table 2: Distribution across the four severity categories

Despite these differences, we use previous results as an indicative starting point. In earlier studies, severity was sometimes predicted using four categories (1 – Unharmed, 2 – Fatal, 3 – Hospitalized, 4 – Minor injury), whereas we consider a binary classification task (severe vs non-severe). Therefore, we regroup the categories from previous work to match our setting.

Specifically, we identified two relevant Kaggle notebooks: one predicts a specific type of accident severity, and another performs a binary classification similar to our task. The results of these studies, along with different machine learning algorithms, are summarized in the table below. The notebooks can be accessed at the following links:

- Hossammaly (Kaggle, 2020): "Predict Severity of Accidents". `https://www.kaggle.com/code/hossammaly/predict-severity-of-accidents`
- Mohamed Elfergouch (Kaggle, 2022): "Predict an Accident in French". `https://www.kaggle.com/code/mohamedelfergouch/predict-an-accident-in-french`

3.2.4 Baseline results from previous work

After adapting the target variable, we reproduced the results from two existing notebooks.

Note: All the changes made in both notebooks can be identified by the key phrase: "CHANGE HERE".

In the first notebook, we obtained the following performance metrics:

Model	**Accuracy**	**Precision**	**Recall**	**F1-score**
Logistic Regression	0.792	0.458	0.002	0.005

Table 3: Results from the first notebook

In the second notebook, using a Random Forest model, we obtained:

Model	**Accuracy**	**Precision**	**Recall**	**F1-score**
Random Forest	0.711	0.694	0.612	0.651

Table 4: Results from the second notebook

3.3 Preprocessing and Data Cleaning

To keep the work organized and clear, we created four separate Python notebooks to process the data and select the most relevant features for our goal of binary classification. The overall structure and relationships between the tables are as follows:

We will examine each type of table across the different years and select the features most relevant for predicting accident severity.

3.3.1 Study of the *Usagers* Table.

The *Usagers* table contains information about the people involved in the accident. Initially, it includes the following attributes:

- `Num_Acc`
- `id_usager`
- `id_vehicule`
- `num_Veh`
- `place`
- `catu`
- `grav`
- `sexe`

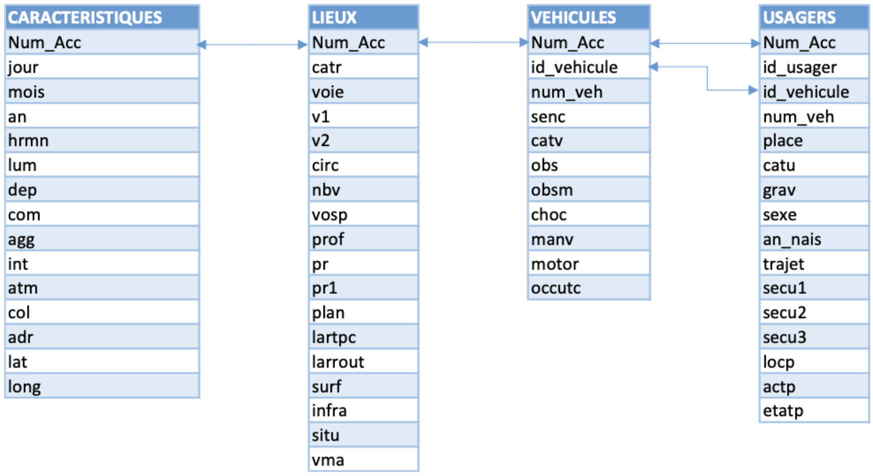


Figure 1: Joining of tables

- `An_nais`
- `trajet`
- `secu1, secu2, secu3`
- `locp`
- `actp`
- `etatp`

A detailed description of these variables is provided in Appendix A.

After ensuring that the data types were correct and replacing all dataset-specific codes for missing values with pandas missing values, we examined the data to determine the proportion of missing values in each category. Here are the results:

Note: Since we have the "id_vehicule" needed to link to the vehicle tables, we can drop the entire "num_veh" column. Additionally, as we do not need to identify each person individually in our analysis (we only require the features, which can be obtained by joining on "Num_Acc" and "id_vehicule"), we can also drop the "id_usager" column.

Feature	**Missing (%)**
secu3	97.31
etatp	92.19
locp	45.91
actp	42.07
secu2	40.59
trajet	27.77
an_nais	1.49
sexe	1.43
secu1	1.34
grav	0.06
Num_Acc	0.00
id_vehicule	0.00
place	0.00
catu	0.00

Table 5: Proportion of missing values per feature for "usagers"

As we can see, the variable `etatp` is missing in more than 90% of cases. Moreover, this feature is only relevant for pedestrians. Due to its very high proportion of missing values, we decided not to use it.

The variables `locp` and `actp` (respectively "Pedestrian location" and "Pedestrian action") also contain a large proportion of missing values (around 40–45%). Since these features are only defined for pedestrians, we did not include them when considering all categories of users (drivers and passengers). However, they were retained along with `etatp` for analyses focusing exclusively on pedestrians.

Finally, the `trajet` ("Trip purpose") variable contains approximately 30% missing values. We performed a simple imputation by replacing missing values with 0, corresponding to the "not reported"category. The results were not different from completely ignoring the feature, so for the sake of saving time, we chose to omit it.

Next, some missing values still remain. For all features except `secu_n` ("Safety measures"), we will simply remove the rows with missing data, which represent at most 1.44%.
For `secu_n`, since these represent safety measures taken, we represent them differently for our analysis. Each entry can include multiple categories of safety measures in no specific order, so overall, the safety measures are represented as a set of these categories. To handle this, we one-hot encoded all the categories. We also removed `gants-airbag` and kept `gants` and `airbag` separately, since they contain all the necessary information.

Then, we transformed the `grav` ("Severity") column into `grav_bin`. Since we decided to perform a binary classification to predict whether the victim is severely injured or not, we grouped categories 1 (*Unharmed*) and 4 (*Slightly injured*) into 0, meaning not heavily injured, and categories 2 (*Killed*) and 3 (*Hospitalized injured*) into 1, meaning heavily injured.

Here is the final set of features obtained from this set of tables:

```
features = ['place', 'catu', 'sexe', 'age', 'aucun', 'ceinture',
            'casque', 'dispositif_enfant', 'gilet_reflechissant',
            'airbag', 'gants', 'autre', 'non determinable']
```

Next, we study the distribution of variables.

In the current dataset, we have among continuous variables only the age, so we decided to start by it. We saw its distribution and noticed some weird extreme values, such as age 120.

Measure	Value
mean	38.60
std	19.01
min	0.00
25%	23.00
50%	35.00
75%	52.00
max	120.00

Table 6: Age Statistics

Then we looked at how many ages we actually have in our dataset that are above 100 and found many of these, and wierdly 133 of people with age 119 ! we found that strange, so we have decided to remove all rows with age above 100. Once we cleaned this, we have this quite good looking distribution of ages :

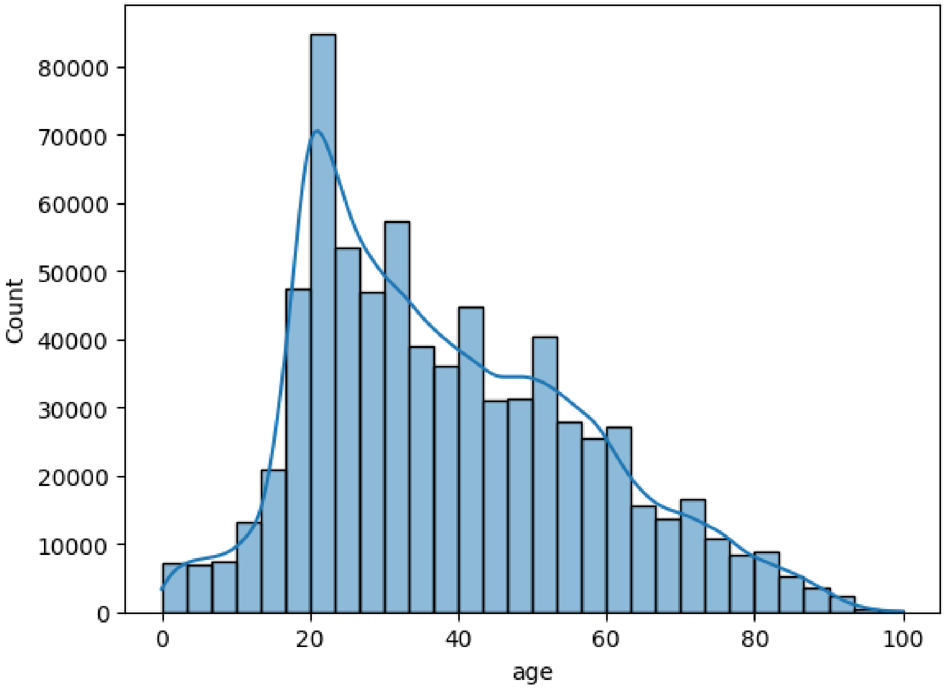


Figure 2: Age distribution

Most of the victims are people of age between 20 and 50, which looks legitimate.
We proceeded with categorical features.
First, we examined the proportion of our target labels:

Target Label	**Proportion**
Not severly Injured	81.69%
Seevrly Injured	18.31%

Table 7: Target distribution

For a more detailed view of the proportions of severe and not severe accidents for each feature category, please refer to the appendix B

Note: Our class distribution is imbalanced. To address this, we apply various techniques such as over- and under-sampling, and ensure that both the cross-validation and train-test splits are stratified. In this context, we rely primarily on the F1-score, as it provides a more relevant and consistent evaluation metric for imbalanced classification tasks.

We have also plotted the correlation matrix to see relations between different features.

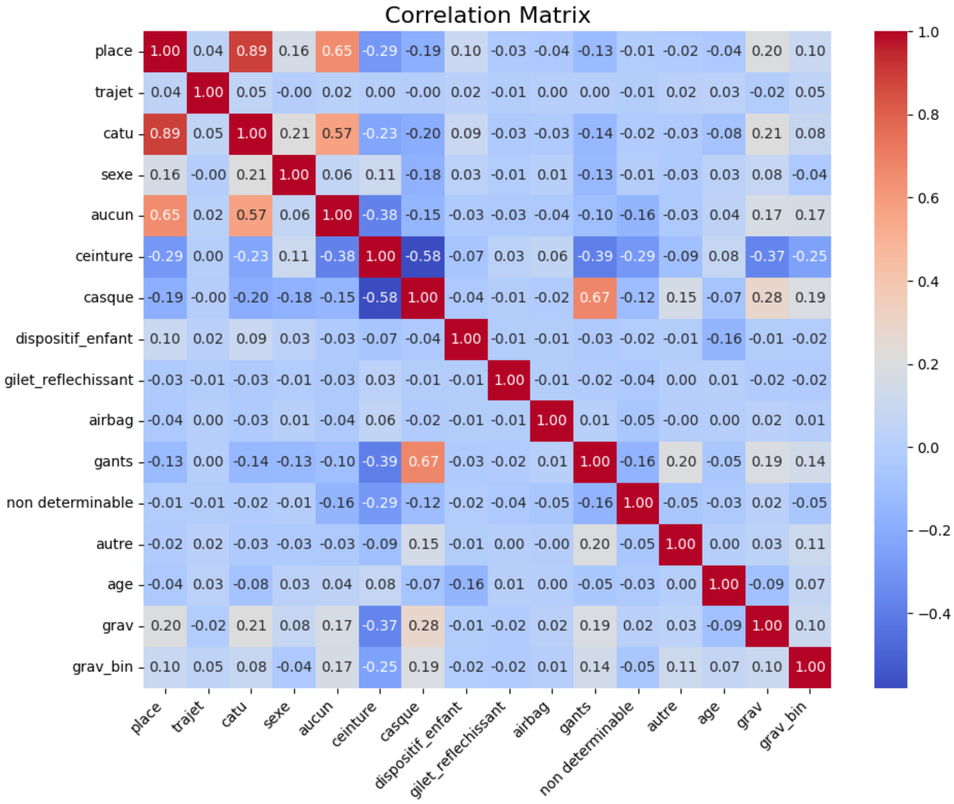


Figure 3: Correlation matrix of the "usagers" set of features

Here are the key insights:

- **Seat position (`place`) and user type (`catu`)** are almost perfectly linked: knowing where someone sat provides strong information about the type of road user.
- **Seatbelt use (`ceinture`) and helmet use (`casque`)** correspond to distinct populations: car occupants wear seatbelts, while motorcyclists wear helmets, with virtually no overlap.
- **Helmet use (`casque`) and gloves (`gants`)** are strongly associated: motorcyclists tend to wear both together, making these two protective items highly correlated.
- Regarding the target variable `grav_bin`, `seatbelt` shows the strongest negative correlation ($r = -0.25$), suggesting that wearing a seatbelt is associated with a lower likelihood of severe injury. Interestingly, `casque` and `gants` exhibit positive correlations with severity ($r = 0.19$ and $r = 0.14$ respectively), which may seem counterintuitive at first glance. However, this likely reflects the fact that these protective items are predominantly worn by motorcyclists and cyclists, whose accidents tend to be more fatal than those involving car occupants, regardless of the protection worn.

Once our dataset containing user-related features is cleaned, we apply several machine learning models to assess how well accident severity can be predicted using only this information. Since accuracy can be misleading in the presence of class imbalance, we rely on metrics such as precision, recall, and F1-score, focusing on the performance of the minority class (severe accidents). We ran the following algorithms: **Logistic Regression**, **Decision Tree**, and **Gradient Boosting**.

Model	**Class 0 (Non-severe)**			**Class 1 (Severe)**		
	Precision	**Recall**	**F1**	**Precision**	**Recall**	**F1**
Logistic Regression	0.89	0.72	0.80	0.33	0.61	0.43
Decision Tree	0.90	0.71	0.79	0.32	0.63	0.43
Gradient Boosting	0.90	0.71	0.79	0.33	0.65	0.44

Table 8: Cross-validation results with oversampling - user-related features only (metrics reported for both classes)

Model	**Class 0 (Non-severe)**			**Class 1 (Severe)**		
	Precision	**Recall**	**F1**	**Precision**	**Recall**	**F1**
Logistic Regression	0.89	0.72	0.80	0.33	0.61	0.43
Decision Tree	0.89	0.70	0.79	0.33	0.63	0.43
Gradient Boosting	0.90	0.71	0.79	0.33	0.65	0.44

Table 9: Cross-validation results with undersampling - user-related features only (metrics reported for both classes)

Gradient Boosting achieves the best performance for the minority class, detecting the greatest number of severe accidents among all candidates with a recall of 0.65 and an F1-score of 0.44, outperforming Logistic Regression and Decision Tree, both yielding an F1-score of 0.43. Oversampling and undersampling produce nearly identical results across all models, suggesting that the user-related features carry a stable but limited discriminative signal for predicting accident severity. Gradient Boosting is therefore selected, and its evaluation on the held-out test set yields the following results:

Class	**Precision**	**Recall**	**F1-score**
0 (Non-severe)	0.90	0.71	0.79
1 (Severe)	0.33	0.64	0.44

Table 10: Classification report - Gradient Boosting (user-related features)

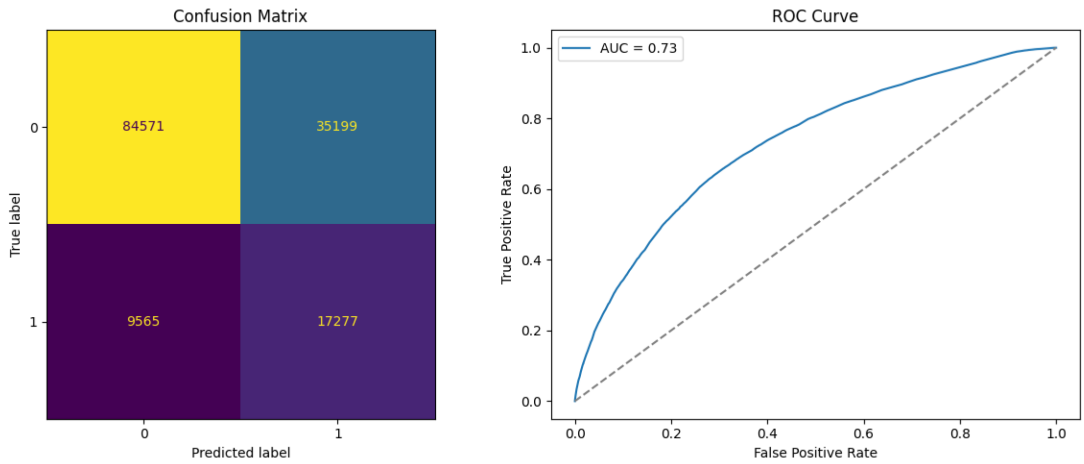


Figure 4: Evaluation of Gradient Boosting - user-related features

Gradient Boosting correctly identifies 64% of severe accidents (recall = 0.64) but with low precision (precision = 0.33), reflecting the class imbalance, while performing well on non-severe cases (F1-score = 0.79), with an AUC of 0.73 suggesting a weak overall discriminative ability for user-related features.

For the pedestrian-specific analysis, we retained only the features directly related to pedestrians:

`['sexe', 'locp', 'actp', 'etatp', 'age', 'aucun', 'gilet_reflechissant', 'autre', 'non_determinable']`.

We also verified the coherence between `place` and `catu`: pedestrians should have `place` = 10 (not applicable) and `catu` = 3 (pedestrian). No inconsistencies were found.

As done previously, we applied several machine learning algorithms using cross-validation with both oversampling and undersampling strategies. The cross-validation results are presented in the tables below :

Model	**Class 0 (Non-severe)**			**Class 1 (Severe)**		
	Precision	**Recall**	**F1**	**Precision**	**Recall**	**F1**
Logistic Regression	0.78	0.62	0.69	0.45	0.64	0.53
Decision Tree	0.74	0.67	0.70	0.43	0.50	0.46
Gradient Boosting	0.78	0.68	0.72	0.47	0.59	0.53

Table 11: Cross-validation results with oversampling - `usagers` features only

Model	**Class 0 (Non-severe)**			**Class 1 (Severe)**		
	Precision	**Recall**	**F1**	**Precision**	**Recall**	**F1**
Logistic Regression	0.78	0.62	0.69	0.45	0.64	0.53
Decision Tree	0.74	0.65	0.69	0.42	0.53	0.47
Gradient Boosting	0.77	0.68	0.72	0.47	0.59	0.52

Table 12: Cross-validation results with undersampling - `usagers` features only

Gradient Boosting is selected as the best model based on the overall balance between both classes. Although Logistic Regression achieves a slightly higher recall for the minority class (0.64 vs 0.59), Gradient Boosting outperforms it on the majority class and achieves a better F1-score overall, making it the most balanced and reliable candidate. Its evaluation on the held-out test set yields the following results:

Class	**Precision**	**Recall**	**F1-score**
0 (Non-severe)	0.78	0.68	0.72
1 (Severe)	0.47	0.60	0.53

Table 13: Classification report - Gradient Boosting (`usagers` features)

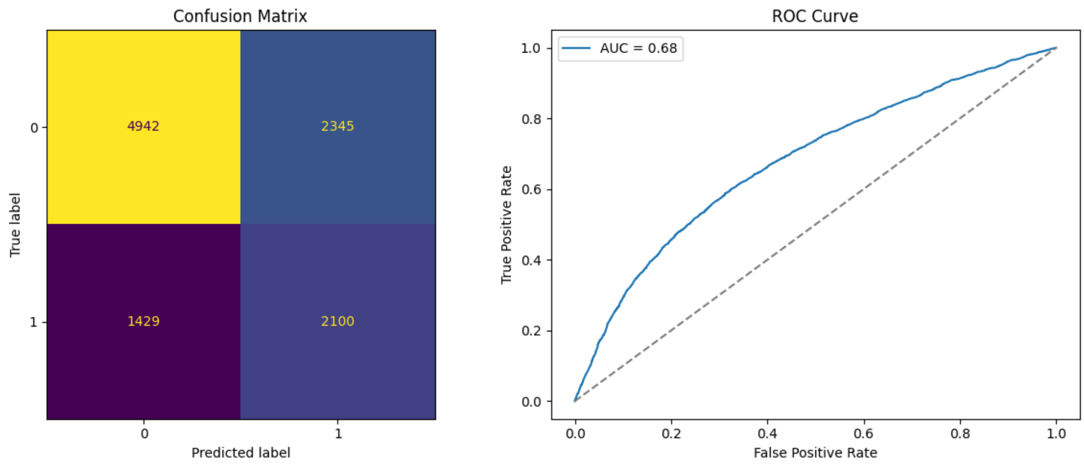


Figure 5: Evaluation of Gradient Boosting - pedestrian features

Gradient Boosting correctly identifies 60% of severe accidents (recall = 0.60) with a more balanced precision (precision = 0.47), yielding an F1-score of 0.53 for the severe class, while maintaining acceptable performance on non-severe cases (F1-score = 0.72), with an AUC of 0.68 indicating a moderate discriminative ability for pedestrian-specific features.

3.3.2 Study of the *Caract* Table.

The *Caract* table contains information about the general circumstances of the accident. Initially, it includes the following attributes:

- `Num_Acc`
- `jour`
- `mois`
- `an`
- `hrmn`
- `lum`
- `dep`
- `com`
- `agg`
- `int`
- `atm`
- `col`
- `adr`
- `lat`
- `long`

For a detailed description of each feature, please refer to Appendix C.

By following similar steps to the first set of feature cleaning, we first ensured that the data types were correct and studied the proportion of missing values for each feature. In this case, we obtained the following results:

Note: In this case, we also immediately dropped several columns: "adr", "dep", "com", and "an". First, we observed that latitude and longitude are always available, providing a more precise and usable representation of the accident location. Therefore, we removed "adr", "dep", and "com". Additionally, we judged that the year of the accident would not be useful for our analysis, as we are more focused on the time of year and the hour of the day.

Note: As the results are rounded to two decimal places, we also reported the exact number of missing values. We obtained the following: all features are complete except for `lum` (9 missing values), `int` (14), `atm` (26), `col` (1619), and `long` (1).

Feature	**Missing (%)**
Num_Acc	0.00
jour	0.00
mois	0.00
hrmn	0.00
lum	0.00
agg	0.00
int	0.00
atm	0.01
col	0.49
lat	0.00
long	0.00

Table 14: Proportion of missing values per feature for "caract"

In this case the number of missing values per feature is quite small, so we have decided to keep all these columns and just to drop the rows with missing values.

Among the continuous variables, we have `lat` and `long`. Their initial distributions were as follows:

Measure	**lat**	**long**
mean	44.34	2.17
std	12.25	19.09
min	-61.42	-178.15
25%	44.74	1.28
50%	47.82	2.39
75%	48.86	4.79
max	63.56	174.02

Table 15: Latitude and Longitude Statistics

We observed some accidents with coordinates outside the expected range for metropolitan France, which may correspond to non-metropolitan regions or erroneous entries. To ensure geographic consistency and facilitate modeling, we filtered the dataset to include only accidents occurring within metropolitan France, defined as latitude between 41° and 51° and longitude between -5° and 10°. In fact, only approximately 5.5% of entries fall outside these bounds.

Once done, we plotted the coordinates, and obtained the following representation.

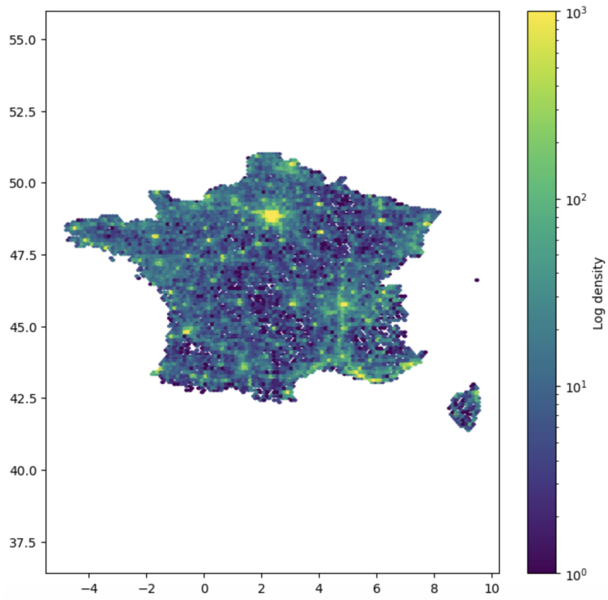


Figure 6: Density distribution in Metropolitain area of France

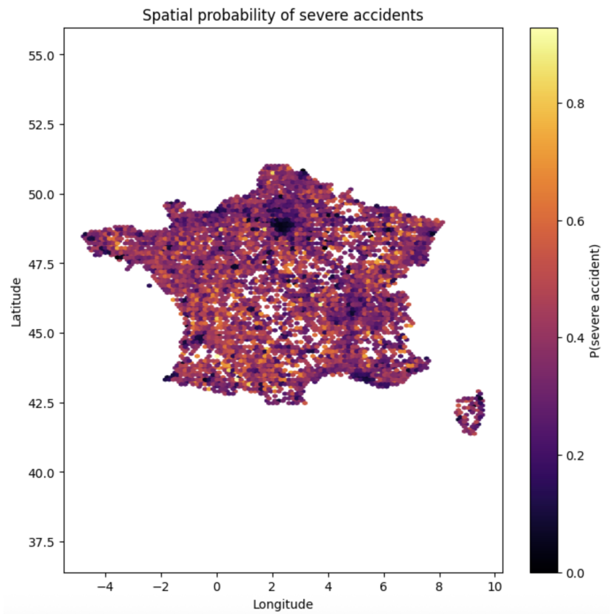


Figure 7: Spatial probability distribution in Metropolitain area of France

We can see that accident density is concentrated in major urban areas, particularly Paris and Lyon, as we can notice clear hotspots on Figure 7. However,the probability of severe accidents is different. Rural and semi-rural regions, despite having fewer accidents overall, show higher rates of severe accidents.

Next, we proceeded with day, month and hrmn (hour minute) numerical discrete features. We obtained the following distribution:

Statistic	Month (mois)	Day (jour)	Time (hrmn_num)
Count	309 533	309 533	309 533
Mean	6.71	15.63	1375
Std	3.36	8.73	551
Min	1	1	0
25%	4	8	950
50%	7	16	1450
75%	10	23	1810
Max	12	31	2359

Table 16: Descriptive statistics for month, day, and time of accidents.

As we can see, the distribution appears reasonable, without any unusual values. However, we decided to transform this feature into a cosine and sine representation of the day of the year. This makes the feature more suitable for prediction, as it better captures cyclical distances compared to a categorical representation. Therefore, we removed "mois", "jour" and "hrmn" and replaced them with "day_sin", "day_cos", "month_sin", "month_cos", "hour_sin", "hour_cos", "weekday_sin", "weekday_cos" and a binary feature "is_weekend".

To study these features, we applied an arctan 2 transformation for better visualization and obtained the following plots :

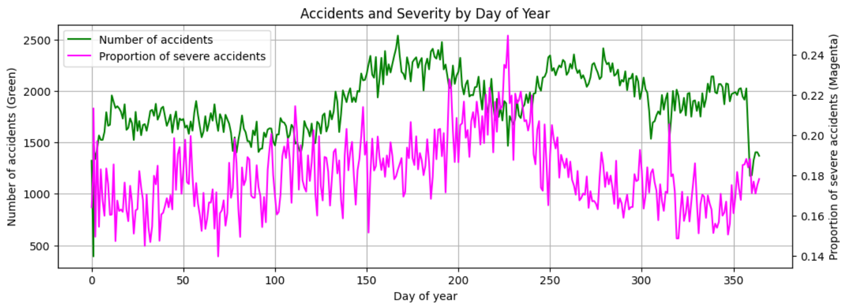


Figure 8: Day of the year by the number and severity of accidents

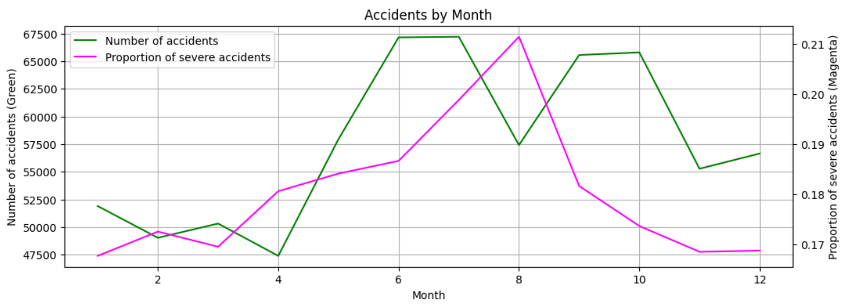


Figure 9: Month by the number and severity of accidents

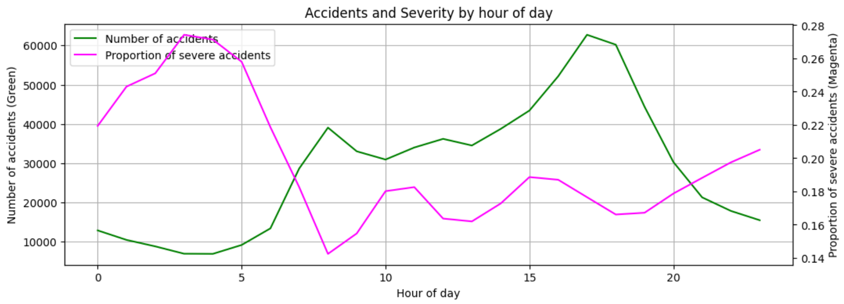


Figure 10: Hour of the day by the number and severity of accidents

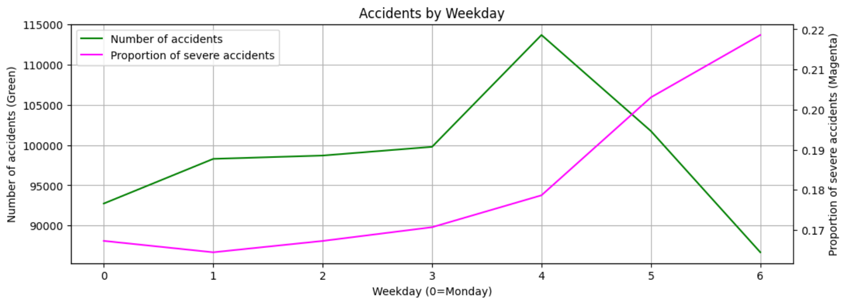


Figure 11: weekday by the number and severity of accidents

The number of accidents (green) and severity (magenta) consistently move in opposite directions across all temporal dimensions. Accidents peak during weekday rush hours, but severity is lowest at these moments. Conversely, nights and weekends show far fewer accidents but a dramatically higher severity rate - likely due to higher speeds, alcohol, and fatigue. The quieter the road, the deadlier the accident.

Then, we proceeded with the categorical features. As before, we displayed the counts per class and the proportion of severe and not severe accidents. Detailed counts and proportions for all categorical features can be found in Appendix D.

And finally, the correlation matrix.

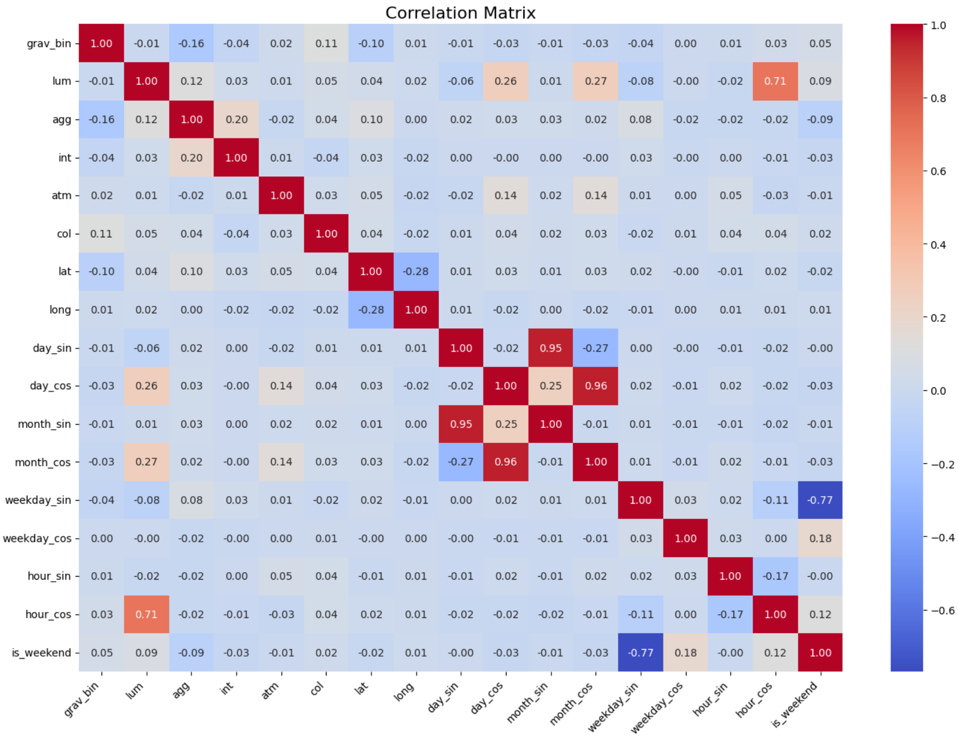


Figure 12: Correlation matrix of the "caract" set of features

The cyclic time features are strongly correlated with each other, as expected: `day_sin/cos`, `month_sin/cos`, and `weekday_sin/cos` form tight pairs with correlations ≥ 0.95. Additionally, `is_weekend` is highly linked to `weekday_sin` ($\rho \approx -0.77$), confirming that these features encode the same temporal information redundantly.

`Latitude` and `longitude` are slightly correlated ($\rho \approx -0.28$), reflecting France's diagonal geography - a spatial signal potentially useful for the model.

Most importantly, `grav_bin` shows very weak correlations with all other features (maximum $\rho \approx 0.16$ with `agg`), confirming that accident severity cannot be predicted by any single variable linearly, and highlighting the need for non-linear models to capture this complex relationship.

Here, we also applied several machine learning algorithms to evaluate how this set of features alone performs in predicting our target. The models were selected using cross-validation with both oversampling and undersampling strategies to account for class imbalance. The results are presented in the tables below.

Note: The final set of features retained from the `caract` table is the following: `['lum', 'agg', 'int', 'atm', 'col', 'lat', 'long', 'day_sin', 'day_cos', 'hour_sin', 'hour_cos', 'month_sin', 'month_cos', 'weekday_sin', 'weekday_cos']`.

Model	**Class 0 (Non-severe)**			**Class 1 (Severe)**		
	Precision	**Recall**	**F1**	**Precision**	**Recall**	**F1**
Logistic Regression	0.90	0.67	0.77	0.31	0.66	0.42
Decision Tree	0.85	0.81	0.83	0.29	0.35	0.32
Gradient Boosting	0.91	0.67	0.77	0.32	0.72	0.45

Table 17: Cross-validation results with oversampling - `caract` features only

Model	Class 0 (Non-severe)			Class 1 (Severe)		
	Precision	Recall	F1	Precision	Recall	F1
Logistic Regression	0.90	0.67	0.77	0.31	0.66	0.42
Decision Tree	0.87	0.65	0.75	0.27	0.57	0.37
Gradient Boosting	0.91	0.67	0.77	0.32	0.72	0.45

Table 18: Cross-validation results with undersampling - `caract` features only

Gradient Boosting achieves the best performance for the minority class, detecting the greatest number of severe accidents with a recall of 0.72 and an F1-score of 0.45. Notably, Logistic Regression and Gradient Boosting produce identical results under undersampling, suggesting that the decision boundary learned from `caract` features is relatively stable, whereas the Decision Tree performs noticeably worse, with an F1-score dropping to 0.37 for class 1. Gradient Boosting is therefore selected, and its evaluation on the held-out test set yields the following results:

Class	Precision	Recall	F1-score
0 (Non-severe)	0.92	0.67	0.77
1 (Severe)	0.32	0.72	0.45

Table 19: Classification report - Gradient Boosting (`caract` features)

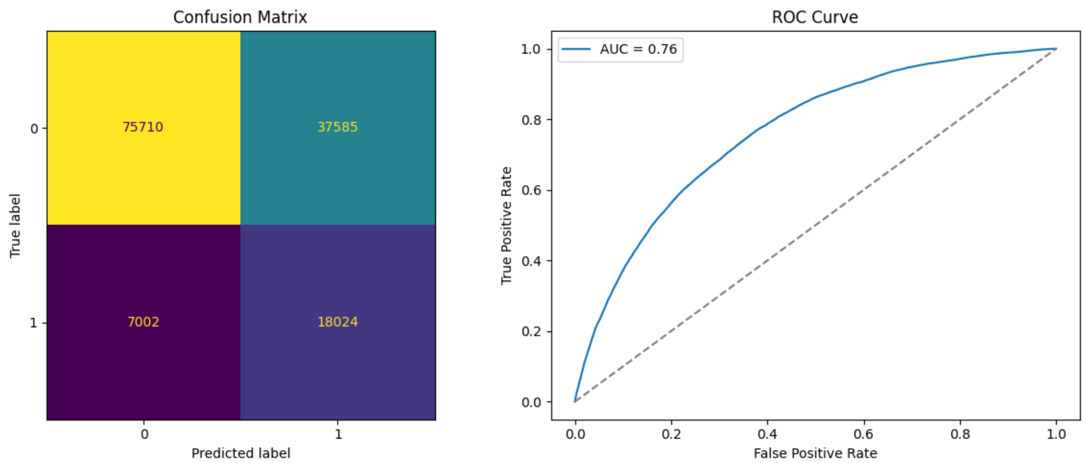


Figure 13: Evaluation of Gradient Boosting - `caract` features

Gradient Boosting, evaluated on the held-out test set, correctly identifies 72% of severe accidents (recall = 0.72) but with low precision (precision = 0.32), reflecting the class imbalance, while performing well on non-severe cases (F1-score = 0.77), with an AUC of 0.76 confirming a reasonable overall discriminative ability.

3.3.3 Study of the *Lieux* Table.

The *Lieux* table contains information describing the main location of the accident, even if it occurred at an intersection. Initially, it includes the following attributes:

- `Num_Acc`
- `catr`
- `voie`
- `V1`
- `V2`
- `circ`
- `nbv`
- `vosp`
- `prof`
- `pr`
- `pr1`
- `plan`
- `lartpc`
- `larrout`
- `surf`
- `infra`
- `situ`
- `vma`

For more details on the description of each attribute, see Appendix E.

We proceed in the same manner: correcting attribute types and handling missing values. We obtain the following percentages for each category:

Feature	**Missing (%)**
Num_Acc	0.00
catr	0.00
voie	10.76
v1	18.58
v2	91.13
circ	5.97
nbv	1.07
vosp	0.50
prof	0.01
pr	7.04
pr1	7.11
plan	0.01
lartpc	99.82
larrout	86.69
surf	0.02
infra	0.94
situ	0.05
vma	1.67

Table 20: Proportion of missing values per feature for "lieux"

We decided to drop the following columns: `'voie', 'v1', 'v2', 'pr', 'pr1', 'lartpc', 'larrout'`, as they contain a large number of missing values and are not relevant for our prediction task.

Specifically, the variables `'voie', 'v1', 'v2'`, which describe the road identification (route number and its variants such as "bis" or "ter") are mainly administrative and don't provide meaningful information about the conditions of the accident or its severity. The variables `'pr'` and `'pr1'`, which correspond to precise geographic positioning along a road (reference point and distance), these locate the accident, but don't describe factors probably influencing severity. Finally, `'lartpc'` (width of the central median) and `'larrout'` (width of the roadway) could theoretically be linked to accident severity. But in our case, their values are missing in more than 86% of cases. Including them would either reduce the dataset size significantly or require complicated imputation, which could introduce bias. So, we decided to remove them.

Next we proceed with feature analysis for the remaining ones :
We start with the maximum authorized speed (`vma`), a numerical discrete variable. The initial distribution reveals two anomalies: a minimum value of 0 and a maximum of 901, both clearly inconsistent with real-world speed limits.

Measure	**Value**
Mean	61.05
Std	24.05
Min	0.00
25%	50.00
50%	50.00
75%	80.00
Max	901.00

Table 21: Descriptive statistics of `vma` before preprocessing

Based on the valid speed limits defined by the French highway code, we define the set of acceptable values as:

$$\mathcal{V} = \{5, 10, 15, 20, 30, 50, 70, 80, 90, 110, 130\}$$

Any value outside $\mathcal{V}$, as well as missing values (`NaN`), were treated as invalid and imputed with the median value of `vma` within the corresponding road category (`catr`) group. This strategy shows the strong correlation between `vma` and `catr`, ensuring that imputed values remain consistent with the typical speed restrictions observed for each road category.

The `nbv` feature (number of traffic lanes) takes integer values between 0 and 12, with 2.97% missing values. However, since every road must have at least one traffic lane, a value of 0 is physically impossible and is therefore treated as a missing value. Furthermore, values strictly greater than 9 are considered erroneous, as roads with more than 9 lanes are extremely rare in practice. Combined, missing and invalid values account for 4.06% of the data (3.71% are missing and 0.35% - others), which is a sufficiently small proportion to justify imputation without significantly affecting the overall distribution.

Given the moderate correlation between `nbv` and `circ` (type of traffic flow), the imputation was performed using the median of `nbv` within each `circ` group, ensuring that the imputed values are consistent with the type of traffic flow observed on the road.

Given that the proportion of missing entries of categorical features was very low (fewer than 25 000 instances out of the entire dataset), imputation was performed by replacing each missing value with the most frequent value of the corresponding feature. This choice is unlikely to significantly affect the results given the negligible proportion of affected rows.

Since the previously discussed numerical discrete features contain a relatively small number of possible values, they are included alongside the categorical features in the analysis table Appendix F, which displays the number of instances per class and the proportion of target values per class.

And finally, the correlation matrix.

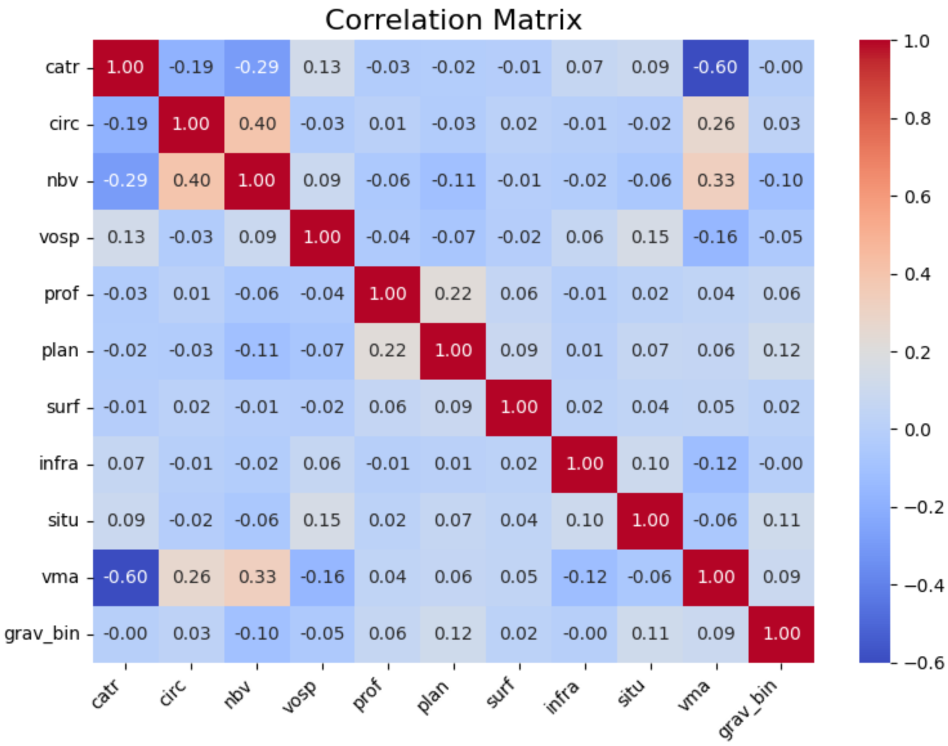


Figure 14: Correlation matrix of the "lieux" set of features

The correlation matrix reveals several notable relationships. The strongest correlation is observed between `catr` and `vma` ($r = -0.60$). A moderate correlation exists between `circ` and `nbv` ($r = 0.40$), which is consistent with the imputation strategy adopted for `nbv`. Additionally, `prof` (The road's incline at the accident location) and `plan` (The road's horizontal alignment) are slightly correlated ($r = 0.22$), as both describe the geometric profile of the road. Finally, all correlations with the target variable `grav_bin` remain very low (below 0.12), suggesting that road infrastructure features alone carry limited discriminative power for predicting accident severity.

Here, we also applied several machine learning algorithms to evaluate how this set of features alone performs in predicting our target. The results are presented in the tables below.

Note: The final set of features retained from the `lieux` table is the following: `['catr', 'circ', 'nbv', 'vosp', 'prof', 'plan', 'surf', 'infra', 'situ', 'vma']`.

Model	**Class 0 (Non-severe)**			**Class 1 (Severe)**		
	Precision	**Recall**	**F1**	**Precision**	**Recall**	**F1**
Logistic Regression	0.89	0.68	0.77	0.31	0.65	0.42
Decision Tree	0.88	0.73	0.80	0.33	0.58	0.42
Gradient Boosting	0.89	0.74	0.81	0.35	0.59	0.44

Table 22: Cross-validation results with oversampling - `lieux` features only

Model	Class 0 (Non-severe)			Class 1 (Severe)		
	Precision	Recall	F1	Precision	Recall	F1
Logistic Regression	0.89	0.67	0.77	0.32	0.67	0.42
Decision Tree	0.88	0.72	0.79	0.33	0.59	0.42
Gradient Boosting	0.89	0.74	0.81	0.35	0.59	0.44

Table 23: Cross-validation results with undersampling - `lieux` features only

Overall, both sampling strategies yield very similar results. Gradient Boosting consistently achieves the best performance across both settings, with the highest F1-score for the minority class (severe accidents) at 0.44. The results also highlight the inherent difficulty of predicting severe accidents from road infrastructure features alone, as evidenced by the low precision for class 1 across all models.

Gradient Boosting is again selected as the best model, as it detects the greatest number of severe accidents among all candidates, with a recall of 0.59 and an F1-score of 0.44, and its evaluation on the held-out test set yields the following results:

Class	Precision	Recall	F1-score
0 (Non-severe)	0.89	0.74	0.81
1 (Severe)	0.34	0.58	0.43

Table 24: Classification report - Gradient Boosting (`lieux` features)

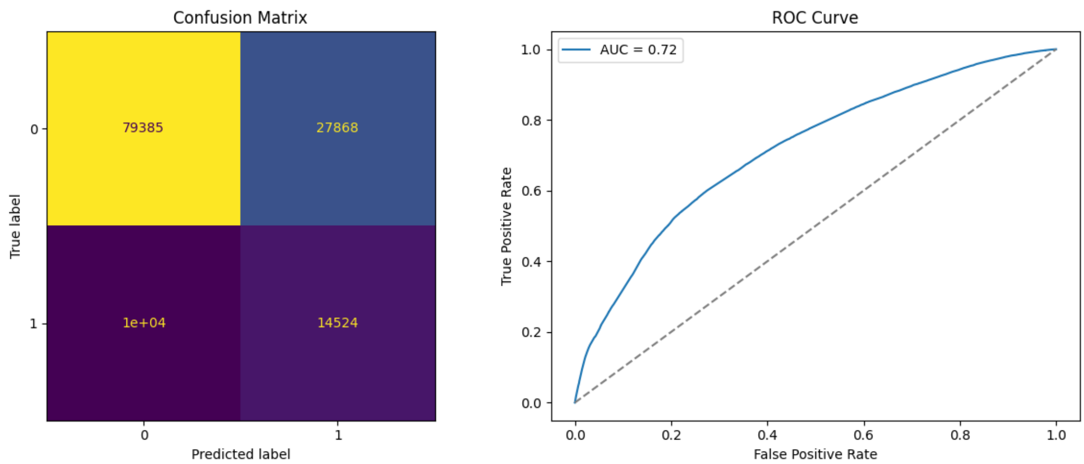


Figure 15: Evaluation of Gradient Boosting - `lieux` features

Gradient Boosting, evaluated on the held-out test set, correctly identifies 58% of severe accidents (recall = 0.58) with low precision (precision = 0.34), yielding an F1-score of 0.42 for the severe class, while performing well on non-severe cases (F1-score = 0.81), with an AUC of 0.72 indicating a moderate discriminative ability for location-related (`lieux`) features.

3.3.4 Study of the *Vehicules* Table.

The *Vehicules* table contains information about the vehicles involved in the accident. Initially, it includes the following attributes:

- `Num_Acc`
- `id_vehicule`
- `Num_Veh`
- `senc`
- `catv`
- `obs`
- `obsm`
- `choc`
- `manv`
- `motor`
- `occutc`

For detailed descriptions of the features in the `Vehicules` dataset, see Appendix G. The following table presents the proportion of missing values for each feature :

Feature	**Missing Values (%)**
`Num_Acc`	0.00%
`id_vehicule`	0.00%
`num_veh`	0.00%
`senc`	0.26%
`catv`	0.00%
`obs`	0.04%
`obsm`	0.05%
`choc`	0.05%
`manv`	0.04%
`motor`	0.22%
`occutc`	99.13%

Table 25: Proportion of missing values in the `vehicules` table

As shown in the table, the proportion of missing values is negligible for most features. The only exception is `occutc`, which represents the number of occupants in public transport and exhibits a missing rate of 99.13%, making it uninformative. This feature was therefore dropped. For all remaining features with missing values, imputation was performed using the most frequent value (mode) of each respective feature. Additionally, `num_veh` was dropped as it does not carry relevant predictive information, given that `id_vehicule` is already available to perform the join with other tables.

For detailed counts and proportions of severe and non-severe accidents across the `Vehicules` features, see Appendix H.

We also plotted the correlation matrix :

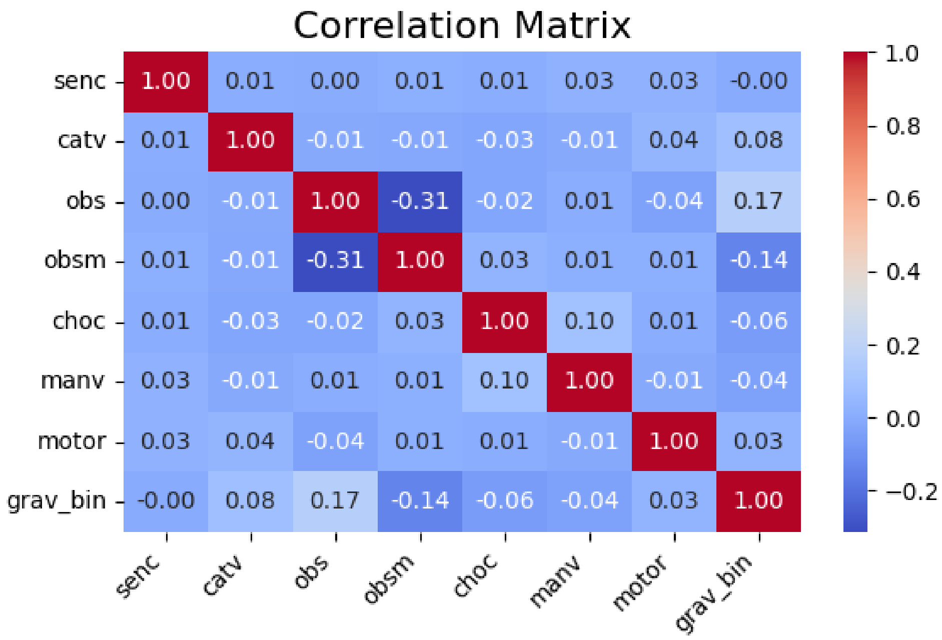


Figure 16: Correlation matrix of the "lieux" set of features

The correlation matrix reveals that `obs` (fixed obstacle hit) and `obsm` (mobile obstacle hit) are negatively correlated ($r = -0.31$), which is expected since a vehicle typically hits either a fixed or a mobile obstacle, rarely both. These two features also exhibit the strongest correlations with the target variable `grav_bin` ($r = 0.17$ and $r = -0.14$ respectively), making them the most informative predictors of accident severity within this feature set. All other features show negligible correlation with `grav_bin`, suggesting that vehicle-related characteristics alone carry limited discriminative power for severity prediction.

As done previously, we applied several machine learning algorithms to evaluate the predictive power of this feature set alone. The final set of features retained from the `vehicules` table is the following: `['catv', 'obs', 'obsm', 'choc', 'manv', 'motor', 'senc']`.

Model	**Class 0 (Non-severe)**			**Class 1 (Severe)**		
	Precision	**Recall**	**F1**	**Precision**	**Recall**	**F1**
Logistic Regression	0.91	0.71	0.79	0.34	0.67	0.45
Decision Tree	0.90	0.72	0.80	0.34	0.65	0.45
Gradient Boosting	0.91	0.71	0.80	0.34	0.67	0.45

Table 26: Cross-validation results with oversampling - `vehicules` features only

Model	**Class 0 (Non-severe)**			**Class 1 (Severe)**		
	Precision	**Recall**	**F1**	**Precision**	**Recall**	**F1**
Logistic Regression	0.91	0.71	0.80	0.34	0.67	0.45
Decision Tree	0.90	0.71	0.79	0.34	0.67	0.45
Gradient Boosting	0.91	0.72	0.80	0.35	0.67	0.45

Table 27: Cross-validation results with undersampling - `vehicules` features only

All three models yield very similar results across both sampling strategies. Gradient Boosting Gradient Boosting is selected as the best model, as it achieves the best overall balance between both classes. While all models perform similarly on the minority class, Gradient Boosting significantly outperforms the others on the majority class, making it the most reliable candidate for this feature set with a recall of 0.66 and an F1-score of 0.45. Its evaluation on the held-out test set yields the following results:

Class	**Precision**	**Recall**	**F1-score**
0 (Non-severe)	0.90	0.71	0.80
1 (Severe)	0.34	0.66	0.45

Table 28: Classification report - Gradient Boosting (`vehicules` features)

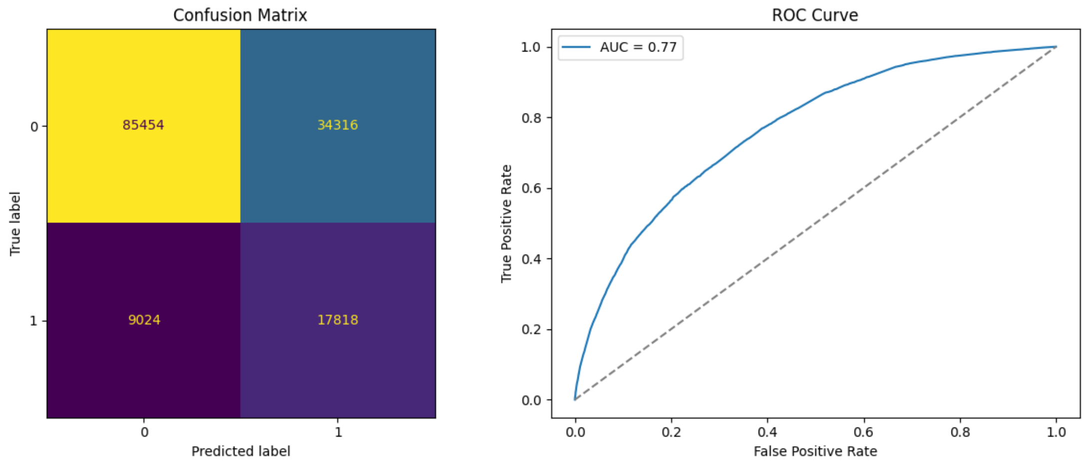


Figure 17: Evaluation of Gradient Boosting - `vehicules` features

Gradient Boosting, evaluated on the held-out test set, correctly identifies 67% of severe accidents (recall = 0.67) with low precision (precision = 0.34), yielding an F1-score of 0.45 for the severe class, while performing well on non-severe cases (F1-score = 0.80), with an AUC of 0.77 indicating a good discriminative ability for vehicle-related (`vehicules`) features.

4 Supervised Learning on the Full Dataset

For the final supervised learning part, we combined the four cleaned tables into one single dataset in order to use all the available features for prediction.

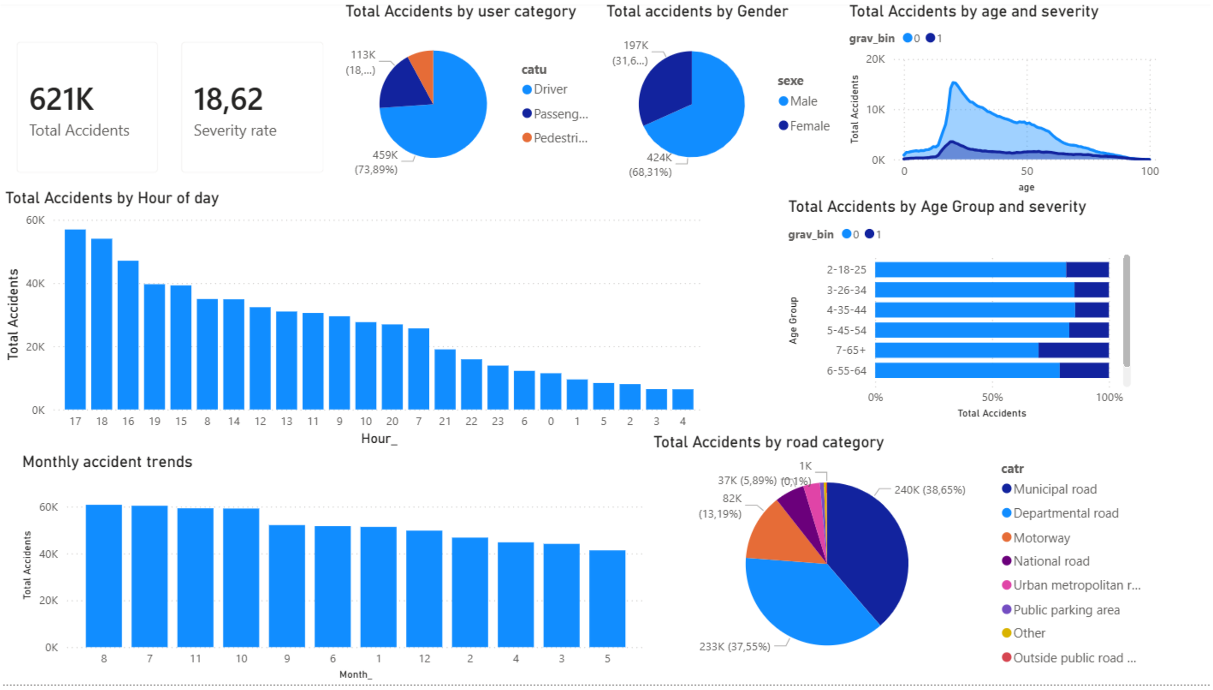


Figure 18: Clean Dataset Preview

In this part, we tested three classification methods:

- Logistic Regression,
- Random Forest,
- Gradient Boosting.

Since our target variable is imbalanced (around 80% of observations in one class and 20% in the other), we also tested different sampling strategies:

- no resampling,
- undersampling,
- oversampling.

Our workflow was the following. For each model, we defined a grid of possible hyperparameter values. Then, we performed a grid search with cross-validation to select the best combination.

An important precision is that, because our dataset is imbalanced, we specified that the main metric used by the grid search to select the best model should be the F1 score, which is by default computed for the positive class. In our case, this positive class corresponds to severely injured or killed individuals. Therefore, we try to improve the balance between precision and recall for this class in order to better detect it.

Also , in this full-dataset analysis, we replaced the single Decision Tree used in the previous smaller analyses by a Random Forest.

In fact, we have read that for larger sets, Random Forests perform better than Decision Trees, which tend to overfit more, while Random Forests combine the predictions of multiple Decision Trees, which helps to address this issue.

4.1 Hyperparameter Choices

4.1.1 Logistic Regression

For Logistic Regression, we used the following hyperparameter grid:

```
param_grid = {
    "model__C": [0.01, 0.1, 1, 10, 100],
    "model__penalty": ["l1", "l2"],
    "model__class_weight": [None, "balanced"]
}
```

We now explain the role of each of these hyperparameters and why these values were chosen.

Parameter `C` The parameter `C` controls the strength of regularization in Logistic Regression. In the regular formula of the objective function, we have:

$$J(\theta) = \text{Loss}(\theta) + \lambda\,\text{Penalty}(\theta) \quad \text{or} \quad J(\theta) = \text{Loss}(\theta) + \frac{1}{C}\,\text{Penalty}(\theta), \quad \text{with } C = \frac{1}{\lambda} \tag{1}$$

Therefore smaller value of `C` gives stronger penalty for overfitting and for larger - weaker.

We chose the values: $\{0.01,\ 0.1,\ 1,\ 10,\ 100\}$ because as we looked on the internet, researchers often use logarithmic range (powers of 10) as it covers a wide range of regularization strengths in an efficient way.

Parameter `penalty` Furthermore, in the previous formula we also have the *Penalty* function, which is specifically this hyper parameter and defines the type of regularization used. We have decided to take the most common choices `l1` (which adds a penalty based on the absolute values of the coefficients) and `l2` (which adds a penalty based on the squared values of the coefficients).

Parameter `class_weight` Finally, we have also decided to consider this hyperparameter, because since our target variable is imbalanced and our main metric is the F1 score of this class, we wanted to test whether giving more weight to the minority class could improve the model's ability to detect it and thus improve the score.

4.1.2 Random Forest

For Random Forest, we used the following hyperparameter grid:

```
param_grid = {
    "model__n_estimators": [100, 200],
    "model__max_depth": [10, None],
    "model__class_weight": [None, "balanced"]
}
```

Parameter `n_estimators` This parameter defines the number of trees in the forest. Higher values for this parameter generally help the model to better generalize.

We chose the values: $\{100,\ 200\}$. Here, 100 is a default value in the `scikit-learn` implementation, and we also wanted to test 200 to see whether the performance improves with more trees.

Parameter `max_depth` This parameter defines the maximum depth of the trees computed. We decided to tune this parameter as well because it directly controls the complexity of each tree: deeper trees can capture more complex patterns, but they can also overfit the training data.

Here, we decided to test 2 different values: `10` and `None`. In the first case, our trees are going to be more constrained, while in the second case, we will see what happens when we let the trees expand until the leaves are pure or until no further split is possible.

Parameter `class_weight` We considered this parameter for the same reason as in the case of Logistic Regression discussed previously.

4.1.3 Gradient Boosting

For Gradient Boosting, we used the following hyperparameter grid:

```
param_grid = {
    "model__n_estimators": [100, 200],
    "model__learning_rate": [0.01, 0.1]
}
```

Parameter `n_estimators` This parameter defines the number of trees used in the boosting process, and here, as in the case of Random Forest, we also decided to take the default number and a double of it for testing.

Parameter `learning_rate` Since Gradient Boosting updates the model by adding trees one by one that correct the errors of the previous ones, one of its main hyperparameters is the learning rate. This parameter controls how much each new tree contributes to the final prediction.

Smaller values lead to more stable learning, but slower learning, while larger values lead to faster learning but increase the risk of overfitting or unstable updates. In our case, we picked $\{0.01,\ 0.1\}$ because these are two standard values that allow us to test smaller and larger learning rates while keeping the grid quite simple.

4.2 Results of Hyperparameter Tuning

After running all trainings, we obtained the following results.

4.2.1 Best parameters found

Table 29: Best hyperparameters found for each model and sampling method

Model \ Sampling method	**Nothing (initial proportion)**	**Undersampling**	**Oversampling**
Logistic Regression	`C = 1, class_weight = balanced, penalty = l1`	`C = 10, class_weight = None, penalty = l1`	`C = 1, class_weight = None, penalty = l2`
Gradient Boosting	`learning_rate = 0.1, n_estimators = 200`	`learning_rate = 0.1, n_estimators = 200`	`learning_rate = 0.1, n_estimators = 200`
Random Forest	`class_weight = balanced, max_depth = 10, n_estimators = 100`	`class_weight = None, max_depth = None, n_estimators = 200`	`class_weight = None, max_depth = None, n_estimators = 200`

First, we can notice that in almost all methods hyperparameters change depending on the sampling method, which shows that the model configuration is sensitive to the class distribution in the training set. Specifically, whenever the `class_weight` parameter can be set to `balanced` (in unbalanced proportion), it is always chosen, which is logical since our main metric is the F1 score of the minority class.

Only, the best parameters for Gradient Boosting remain the same across the sampling methods tested.

4.2.2 Best cross-validated F1 score for minority class

From these results, we can see that all methods give approximately similar results around 50%. However, we can notice that Logistic Regression has roughly the same performance across the three sampling techniques, while for the tree-based methods, the F1 score increases when using more balanced proportions and more samples. This may suggest that a linear model is not the most suitable choice for predicting our target.

We can also see that the worst cross-validation F1 score is obtained with Gradient Boosting on the original (unbalanced) data. This looks justifiable, as it is the only method among the three for which we did not set the `class_weight` parameter to `balanced`, indicating that this option can be quite useful with imbalanced classes.

Finally, Random Forest achieves the best results with both undersampling and oversampling, reaching almost 60%.

Table 30: Cross-validated minority-class F1 score

Model \ Sampling method	**Nothing (initial proportion)**	**Undersampling**	**Oversampling**
Logistic Regression	0.5464	0.5461	0.5464
Gradient Boosting	0.4872	0.5692	0.5694
Random Forest	0.5531	0.5803	0.5716

4.2.3 Most Useful and Least Useful Features for Target Prediction

We also decided to investigate which features each model considers the most important and most useful for prediction. The results are shown below:

Note: For each model, we considered the version trained with the sampling method that achieved the highest cross-validated F1-score. Therefore, for Gradient Boosting we selected oversampling, for Logistic Regression also oversampling, and for Random Forest, undersampling.

Table 31: Top 5 and Bottom 5 Features by Importance for Each Model

Model	**Top 5 Features (Importance)**	**Bottom 5 Features (Importance)**
Gradient Boosting	ceinture (0.234), lat (0.103), agg (0.099), obs (0.095), obsm (0.071)	prof (0.000), month_cos (0.000), airbag (0.000), weekday_cos (0.000), is_weekend (0.000)
Logistic Regression	catv (19.648), manv (8.703), obs (5.228), obsm (4.391), place (3.552)	day_cos (0.036), weekday_sin (0.023), long (0.010), month_cos (0.009), weekday_cos (0.003)
Random Forest	lat (0.058), long (0.049), age (0.047), catv (0.047), manv (0.045)	is_weekend (0.006), autre (0.0019), airbag (0.0013), dispositif_enfant (0.001), gilet_reflechissant (0.001)

Although the sampling method can cause features to be placed slightly differently or have slightly different importance scores, in general, for a specific ML method (Logistic Regression, Gradient Boosting, or Random Forest), the sets of most important and least important features are almost the same. However, we can see that across different methods, the sets of important features differ.

4.2.4 Results of the best models on test set

Finally, we took the best model with the best hyperparameters found according to the F1-score, which appeared to be Random Forest with undersampling, and tested it on the test set.

Here are the main results we obtained:

Class 0 (Non-severe)			Class 1 (Severe)		
Precision	**Recall**	**F1**	**Precision**	**Recall**	**F1**
0.95	0.77	0.85	0.45	0.83	0.58

Table 32: Results on test set with Random Forest undersampling: metrics

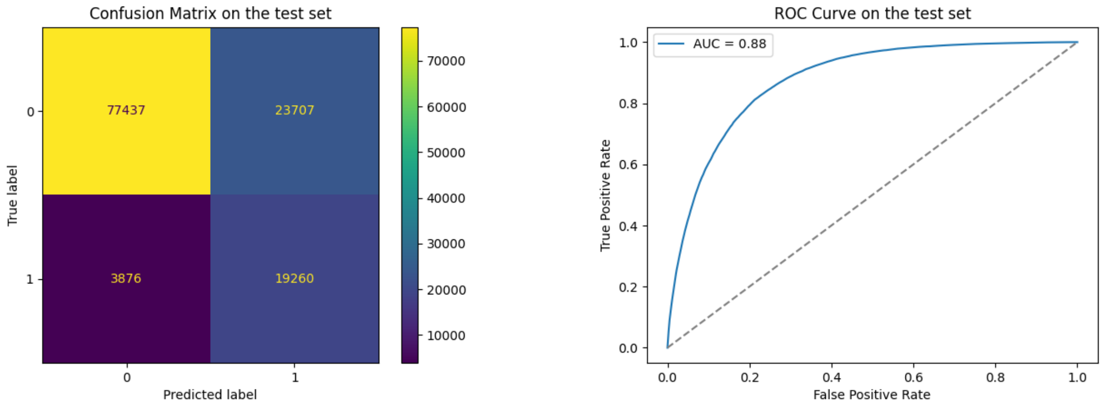


Figure 19: Results on the test set with Random Forest and undersampling: confusion matrix (left) and ROC curve (right)

As we can see from the first table of results, the prediction of the dominant class 0 (non-severe cases) is very good across all metrics. However, for class 1 (severe accidents), we can observe that the lowest metric is precision, with a value of 0.45. This means that among the observations predicted as class 1, only around 45% are actually severe cases; in other words, more than half of these predictions are false alarms. On the other hand, the recall is quite high at 83%, which means that the model is less likely to miss a severe case.

Next, from the confusion matrix, we can also see that the majority class is well identified, while the minority class is less well detected. Although undersampling helped to improve the results, the model still remains biased toward the majority class.

Finally, from the ROC curve, we can see that the area under the curve is relatively strong, with a value of 0.88. This indicates that the model has a good overall ability to distinguish between the two classes. However, in our case, this metric is not the most representative one, since we are working with an imbalanced dataset.

5 Conclusion

In this project, we applied machine learning techniques to predict the severity of road accidents in France using data from 2019 to 2024. Starting from raw tables, we performed thorough preprocessing, feature engineering, and exploratory analysis before building predictive models.

Our analysis showed that each table contributes differently to the prediction task. Accident circumstances (*caract*) provided the strongest signal overall. In the middle, we have user-related features (*usagers*) and vehicle features (*vehicules*), while location features (*lieux*) were the weakest when considered in isolation.

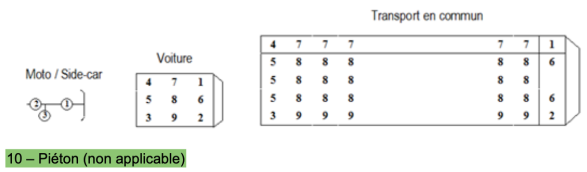
When combining all features into a single dataset, Random Forest with undersampling was found to be the best model, achieving an F1-score of almost 0.60 for the minority class (severe accidents) and an AUC of 0.88 on the test set. Some of the most informative features for the models were seatbelt use, geographic location (latitude/longitude), vehicle category, and the age of the victim.

Finally, predicting severe accidents remains a difficult task, possibly because of the class imbalance. The model still struggles to predict severe cases, which belong to the minority class.

6 Appendix

A Detailed description of the Usagers dataset

Table 33: Description of features in the "Usagers" dataset

Feature	Description
Num_Acc	Accident identification number - Numeric code.
id_usager	Unique user identifier - Numeric code.
id_vehicule	Unique vehicle identifier (same for all users in the same vehicle, including pedestrians linked to that vehicle) - Numeric code.
num_Veh	Vehicle identifier (alphanumeric code used to identify the vehicle for each user, including pedestrians linked to that vehicle).
place	Position of the person in the vehicle at the time of the accident (not applicable for pedestrians). See illustration below: 
catu	Category of the person: 1 – Driver, 2 – Passenger, 3 – Pedestrian.
grav	Severity of injury: 1 - Unharmed 2 – Killed 3 – Hospitalized injury 4 – Minor injury
sexe	Sex of the person: 1 – Male, 2 – Female.
An_nais	Year of birth of the person involved.
trajet	Purpose of travel: -1/0 – Not reported, 1 – Home to work, 2 – Home to school, 3 – Shopping/errands, 4 – Work, 5 – Leisure, 9 – Other.
secu1, secu2, secu3	Safety equipment: -1 – Not reported, 0 – None, 1 – Seatbelt, 2 – Helmet,

Feature	Description
	3 – Child restraint, 4 – Reflective vest, 5 – Airbag, 6 – Gloves, 7 – Gloves + Airbag, 8 – Not determinable, 9 – Other.
`locp`	Pedestrian location: -1 – Not reported, 0 – N/A, 1–2 – On road near crosswalk, 3–4 – On crosswalk w/o/with traffic lights, 5–9 – Sidewalk, shoulder, refuge, unknown, or other.
`actp`	Pedestrian action: -1 – Not reported, 0 – N/A, 1 – Moving with vehicle, 2 – Moving opposite vehicle, 3–6 – Crossing, hidden, playing/running, with animal, 9 – Other, A – Getting in/out of vehicle, B – Unknown.
`etatp`	Pedestrian alone or not: -1 – Not reported, 1 – Alone, 2 – Accompanied, 3 – In a group.

B Detailed proportions of severe and not severe accidents for each category of selected features of "usager"

Feature	Category	Count	Not severe (%)	Severe (%)
aucun	Absent	667 771	83.7	16.3
	Present	65 288	61.2	38.8
ceinture	Absent	295 021	70.0	30.0
	Present	438 038	89.6	10.4
casque	Absent	597 573	85.2	14.8
	Present	135 486	66.3	33.7
dispositif_enfant	Absent	727 533	81.6	18.4
	Present	5 526	89.7	10.3
gilet_reflechissant	Absent	724 763	81.6	18.4
	Present	8 296	90.5	9.5
airbag	Absent	723 642	81.7	18.3
	Present	9 417	79.6	20.4
gants	Absent	662 663	83.5	16.5
	Present	70 396	65.1	34.9
autre	Absent	724 409	82.2	17.8
	Present	8 650	41.4	58.6
non determinable	Absent	582 716	80.7	19.3
	Present	150 343	85.7	14.3
sexe	Female	232 309	83.9	16.1
	Male	500 750	80.7	19.3
catu	Driver	541 998	82.9	17.1
	Passenger	133 759	83.3	16.7
	Pedestrian	57 302	67.1	32.9
place	Driver	541 189	82.8	17.2
	Front left passenger	4 032	86.3	13.7
	Front right passenger	83 917	82.1	17.9
	Other public transport	7 733	78.0	22.0
	Other seat in light vehicle	1 016	85.6	14.4
	Pedestrian (not applicable)	57 302	67.1	32.9
	Public transport – seated	7 136	78.0	22.0
	Public transport – standing	3 237	80.8	19.2
	Rear left passenger	13 176	89.7	10.3
	Rear right passenger	14 321	89.6	10.4

Table 34: Counts and proportions of severely injured and not severely injured for each category of selected features of "usager".

C Detailed description of the Caract dataset

Table 35: Description of features in the "Caract" dataset

Feature	Description
`Num_Acc`	Accident identification number.
`jour`	Day of the accident.
`mois`	Month of the accident.
`an`	Year of the accident.
`hrmn`	Time of the accident (hours and minutes).
`lum`	Lighting conditions: 1 – Daylight, 2 – Twilight or dawn, 3 – Night without public lighting, 4 – Night with public lighting not lit, 5 – Night with public lighting lit.
`dep`	Department code (INSEE code).
`com`	Municipality code (INSEE code: department code + 3 digits).
`agg`	Location type: 1 – Outside built-up area, 2 – Inside built-up area.
`int`	Intersection type: 1 – No intersection, 2 – X intersection, 3 – T intersection, 4 – Y intersection, 5 – Intersection with more than 4 branches, 6 – Roundabout, 7 – Square, 8 – Level crossing, 9 – Other intersection.
`atm`	Weather conditions: -1 – Not reported, 1 – Normal, 2 – Light rain, 3 – Heavy rain, 4 – Snow / hail, 5 – Fog / smoke, 6 – Strong wind / storm, 7 – Dazzling weather (glare), 8 – Overcast, 9 – Other.
`col`	Collision type: -1 – Not reported, 1 – Two vehicles – head-on, 2 – Two vehicles – rear-end, 3 – Two vehicles – side collision, 4 – Three or more vehicles – chain collision, 5 – Three or more vehicles – multiple collisions, 6 – Other collision, 7 – No collision.
`adr`	Postal address (only for accidents in built-up areas).

Feature	Description
`lat`	Latitude.
`long`	Longitude.

D Detailed counts and proportions of features

Feature	Category	Count	Not severe (%)	Severe (%)
`lum`	Daylight	464 483	82.7	17.3
	Night with public lighting lit	104 564	86.5	13.5
	Night with public lighting not lit	6 334	79.2	20.8
	Night without public lighting	70 632	71.4	28.6
	Twilight or dawn	44 879	79.4	20.6
`agg`	Inside built-up area	425 101	86.9	13.1
	Outside built-up area	265 791	73.9	26.1
`int`	Intersection with more than 4 branches	4 141	91.0	9.0
	Level crossing	1 561	78.0	22.0
	No intersection	438 234	79.6	20.4
	Other intersection	29 089	83.2	16.8
	Roundabout	27 274	83.9	16.1
	Square	6 896	92.2	7.8
	T intersection	76 865	85.6	14.4
	X intersection	91 522	87.1	12.9
	Y intersection	15 310	83.7	16.3
`atm`	Dazzling weather (glare)	12 739	73.8	26.2
	Fog / smoke	5 149	71.2	28.8
	Heavy rain	16 293	81.6	18.4
	Light rain	77 462	84.8	15.2
	Normal	543 659	81.9	18.1
	Other	3 142	73.8	26.2
	Overcast	27 651	82.3	17.7
	Snow / hail	2 769	80.8	19.2
	Strong wind / storm	2 028	69.5	30.5
`col`	No collision	42 747	62.3	37.7
	Other collision	167 760	74.4	25.6
	Three or more vehicles – chain collision	46 433	95.3	4.7
	Three or more vehicles – multiple collisions	36 928	86.9	13.1
	Two vehicles – head-on	78 065	72.8	27.2
	Two vehicles – rear-end	98 665	90.5	9.5
	Two vehicles – side collision	220 294	87.1	12.9
`is_weekend`	0.0	502 601	83.0	17.0
	1.0	188 291	79.0	21.0

Table 36: Counts and proportions of severely injured and not severely injured for each category of selected features of "caract".

E Detailed description of features in the "Lieux" dataset

Table 37: Description of features in the "Lieux" dataset

Feature	Description
`Num_Acc`	Accident identification number, identical to the one in the "CARACTERISTIQUES" file, repeated for each vehicle involved in the accident - Numeric code.
`catr`	Road category: 1 – Motorway, 2 – National road, 3 – Departmental road, 4 – Municipal road, 5 – Outside public road network, 6 – Public parking area, 7 – Urban metropolitan roads, 9 – Other.
`voie`	Road number.
`V1`	Numeric index of the road number (e.g., 2 bis, 3 ter).
`V2`	Alphanumeric index letter of the road.
`circ`	Traffic direction: -1 – Not reported, 1 – One-way, 2 – Two-way, 3 – Divided road (separate carriageways), 4 – Variable lanes.
`nbv`	Total number of traffic lanes.
`vosp`	Special lane type: -1 – Not reported, 0 – Not applicable, 1 – Cycle path, 2 – Cycle lane, 3 – Reserved lane.
`prof`	Road profile (slope at accident location): -1 – Not reported, 1 – Flat, 2 – Uphill or downhill slope, 3 – Hilltop, 4 – Bottom of hill.
`pr`	Reference point (PR) number (upstream marker; -1 if not reported).
`pr1`	Distance in meters from the reference point (PR; -1 if not reported).
`plan`	Road layout: -1 – Not reported, 1 – Straight section, 2 – Left curve, 3 – Right curve, 4 – S-shaped curve.
`lartpc`	Width of the central median (in meters), if present.
`larrout`	Width of the roadway used for vehicle traffic (in meters, excluding shoulders, medians, and parking areas).
`surf`	Road surface condition:

Feature	Description
	-1 – Not reported, 1 – Normal, 2 – Wet, 3 – Puddles, 4 – Flooded, 5 – Snow-covered, 6 – Mud, 7 – Icy, 8 – Oil or other slippery substance, 9 – Other.
`infra`	Infrastructure type: -1 – Not reported, 0 – None, 1 – Tunnel, 2 – Bridge, 3 – Ramp or interchange, 4 – Railway crossing, 5 – Designed intersection, 6 – Pedestrian area, 7 – Toll area, 8 – Roadworks, 9 – Other.
`situ`	Accident location: -1 – Not reported, 0 – None, 1 – On roadway, 2 – On emergency lane, 3 – On shoulder, 4 – On sidewalk, 5 – On cycle path, 6 – On other special lane, 8 – Other.
`vma`	Maximum authorized speed at the location and time of the accident.

F Counts and proportions in the "Lieux" dataset

Table 38: Counts and proportions of severely injured and not severely injured for each category of selected features of "lieux".

Feature	Category	Count	Not severe (%)	Severe (%)
`catr`	Departmental road	243 557	72.3	27.7
	Motorway	83 692	89.6	10.4
	Municipal road	252 410	87.2	12.8
	National road	50 739	79.8	20.2
	Other	3 940	76.0	24.0
	Outside public road network	718	71.4	28.6

	Public parking area	5 023	80.2	19.8
	Urban metropolitan roads	20 516	82.9	17.1
`circ`	Divided road	108 379	87.0	13.0
	One-way	115 043	90.6	9.4
	Two-way	433 366	77.2	22.8
	Variable lanes	3 807	85.8	14.2
`nbv`	1.0	59 623	85.6	14.4
	2.0	420 788	77.0	23.0
	3.0	57 955	89.5	10.5
	4.0	84 842	88.4	11.6
	5.0	10 818	87.7	12.3
	6.0	19 010	93.2	6.8
	7.0	1 821	92.8	7.2
	8.0	5 738	95.8	4.2
`vosp`	Cycle lane	15 045	87.4	12.6
	Cycle path	28 122	87.9	12.1
	Not applicable	595 307	80.5	19.5
	Reserved lane	22 121	87.7	12.3
`prof`	Bottom of hill	9 301	74.3	25.7
	Flat	533 462	82.5	17.5
	Hilltop	11 339	74.5	25.5
	Slope	106 493	75.9	24.1
`plan`	Left curve	59 191	71.5	28.5
	Right curve	57 489	71.6	28.4
	S-shaped curve	8 975	67.5	32.5
	Straight section	534 940	83.5	16.5
`surf`	Flooded	350	77.7	22.3
	Icy	2 565	72.3	27.7
	Mud	427	55.5	44.5
	Normal	529 919	81.1	18.9
	Oil or other slippery substance	901	74.6	25.4
	Other	3 011	64.5	35.5
	Puddles	1 036	79.1	20.9
	Snow-covered	1 233	80.5	19.5
	Wet	121 153	82.3	17.7
`infra`	Bridge	12 615	81.0	19.0
	Designed intersection	31 101	82.3	17.7
	None	560 819	80.9	19.1
	Other	23 743	80.1	19.9
	Pedestrian area	5 834	83.9	16.1
	Railway crossing	2 108	79.6	20.4
	Ramp or interchange	9 398	85.3	14.7
	Roadworks	4 977	82.2	17.8
	Toll area	573	73.8	26.2
	Tunnel	9 427	93.2	6.8
`situ`	On cycle path	11 433	89.7	10.3
	On emergency lane	5 761	78.8	21.2
	On other special lane	7 544	86.4	13.6
	On roadway	568 178	83.8	16.2
	On shoulder	38 312	50.1	49.9
	On sidewalk	11 151	71.7	28.3

	Other	18 216	64.8	35.2
vma	5.0	258	85.7	14.3
	10.0	1 182	84.2	15.8
	15.0	371	84.6	15.4
	20.0	1 760	82.3	17.7
	30.0	66 678	88.2	11.8
	50.0	328 409	84.9	15.1
	70.0	53 691	81.3	18.7
	80.0	97 042	60.3	39.7
	90.0	61 227	84.3	15.7
	110.0	30 337	86.8	13.2
	130.0	19 640	79.0	21.0

Table 38: Counts and proportions of severely injured and not severely injured for each category of selected features of "lieux".

G Detailed description of features in the "Vehicules" dataset

Table 39: Description of features in the "Vehicules" dataset

Feature	Description
Num_Acc	Accident identification number, identical to the one in the "CARACTERISTIQUES" file, repeated for each vehicle involved in the accident - Numeric Code.
id_vehicule	Unique vehicle identifier, repeated for each user occupying the vehicle (including pedestrians attached to the vehicle that hit them) – numeric code.
Num_Veh	Vehicle identifier, repeated for each user occupying the vehicle (including pedestrians attached to the vehicle that hit them) – alphanumeric code.
senc	Traffic direction: -1 – Not reported, 0 – Unknown, 1 – Increasing PK or PR or postal address number, 2 – Decreasing PK or PR or postal address number, 3 – No reference point.
catv	Vehicle category: 00 – Undeterminable, 01 – Bicycle, 02 – Moped <50cc, 03 – Microcar (enclosed motorized quadricycle), 04 – Unused reference since 2006 (registered scooter), 05 – Unused reference since 2006 (motorcycle), 06 – Unused reference since 2006 (sidecar), 07 – Passenger car only, 08 – Unused reference since 2006 (car + caravan), 09 – Unused reference since 2006 (car + trailer),

Feature	Description
	10 – Light utility vehicle only 1.5T $\leq$ GVWR $\leq$ 3.5T with or without trailer, 11 – Unused reference since 2006 (LUV (10) + caravan), 12 – Unused reference since 2006 (LUV (10) + trailer), 13 – Heavy goods vehicle only 3.5T $<$ GVWR $\leq$ 7.5T, 14 – Heavy goods vehicle only $>$ 7.5T, 15 – Heavy goods vehicle $>$ 3.5T + trailer, 16 – Road tractor only, 17 – Road tractor + semi-trailer, 18 – Unused reference since 2006 (public transport), 19 – Unused reference since 2006 (tramway), 20 – Special vehicle, 21 – Agricultural tractor, 30 – Scooter $<$ 50cc, 31 – Motorcycle $>$ 50cc and $\leq$ 125cc, 32 – Scooter $>$ 50cc and $\leq$ 125cc, 33 – Motorcycle $>$ 125cc, 34 – Scooter $>$ 125cc, 35 – Light quad $\leq$ 50cc (non-enclosed motorized quadricycle), 36 – Heavy quad $>$ 50cc (non-enclosed motorized quadricycle), 37 – Bus, 38 – Coach, 39 – Train, 40 – Tramway, 41 – 3-wheel motor vehicle $\leq$ 50cc, 42 – 3-wheel motor vehicle $>$ 50cc $\leq$ 125cc, 43 – 3-wheel motor vehicle $>$ 125cc, 50 – Motorized personal transport device, 60 – Non-motorized personal transport device, 80 – Electric-assisted bicycle, 99 – Other vehicle.
`obs`	Fixed obstacle hit: -1 – Not reported, 0 – Not applicable, 1 – Parked vehicle, 2 – Tree, 3 – Metal guardrail, 4 – Concrete guardrail, 5 – Other guardrail, 6 – Building, wall, bridge pier, 7 – Vertical road sign support or emergency call post, 8 – Pole, 9 – Street furniture, 10 – Parapet, 11 – Island, refuge, bollard, 12 – Curb, 13 – Ditch, embankment, rock wall, 14 – Other fixed obstacle on roadway,

Feature	Description
	15 – Other fixed obstacle on sidewalk or shoulder, 16 – Run-off-road without obstacle, 17 – Culvert – culvert head.
`obsm`	Mobile obstacle hit: -1 – Not reported, 0 – None, 1 – Pedestrian, 2 – Vehicle, 4 – Rail vehicle, 5 – Domestic animal, 6 – Wild animal, 9 – Other.
`choc`	Initial point of impact: -1 – Not reported, 0 – None, 1 – Front, 2 – Front right, 3 – Front left, 4 – Rear, 5 – Rear right, 6 – Rear left, 7 – Right side, 8 – Left side, 9 – Multiple impacts (rollovers).
`manv`	Main maneuver before the accident: -1 – Not reported, 0 – Unknown, 1 – No change of direction, 2 – Same direction, same lane, 3 – Between two lanes, 4 – Reversing, 5 – Wrong-way driving, 6 – Crossing the central reservation, 7 – In the bus lane, same direction, 8 – In the bus lane, opposite direction, 9 – Merging, 10 – Making a U-turn on the roadway, 11 – Left (changing lane), 12 – Right (changing lane), 13 – Left (swerving), 14 – Right (swerving), 15 – Left (turning), 16 – Right (turning), 17 – Left (overtaking), 18 – Right (overtaking), 19 – Crossing the roadway, 20 – Parking maneuver, 21 – Evasive maneuver, 22 – Door opening, 23 – Stopped (excluding parking),

Feature	Description
	24 – Parked with occupants, 25 – Driving on sidewalk, 26 – Other maneuvers.
`motor`	Vehicle engine type: -1 – Not reported, 0 – Unknown, 1 – Hydrocarbons, 2 – Hybrid electric, 3 – Electric, 4 – Hydrogen, 5 – Human-powered, 6 – Other.
`occutc`	Number of occupants in public transport.

H Counts and proportions in the "Vehicules" dataset

Table 40: Counts and proportions of severely injured and not severely injured for each category of selected features of `vehicules`.

Feature	Category	Count	Not severe (%)	Severe (%)
`catv`	3-wheel motor vehicle $\leq$ 50cc	108	76.9	23.1
	3-wheel motor vehicle $>$ 125cc	3 762	79.9	20.1
	3-wheel motor vehicle $>$ 50cc $\leq$ 125cc	153	84.3	15.7
	Agricultural tractor	1 664	80.3	19.7
	Bicycle	32 707	72.7	27.3
	Bus	7 202	93.6	6.4
	Coach	2 718	93.6	6.4
	Electric-assisted bicycle	3 641	73.4	26.6
	Heavy goods vehicle $>$ 3.5T + trailer	5 891	90.3	9.7
	Heavy goods vehicle only 3.5T $<$ GVWR $\leq$ 7.5T	2 650	91.1	8.9
	Heavy goods vehicle only $>$ 7.5T	4 855	88.9	11.1
	Heavy quad $>$ 50cc (non-enclosed motorized quadricycle)	1 354	42.6	57.4
	Light quad $\leq$ 50cc (non-enclosed motorized quadricycle)	170	52.4	47.6
	Light utility vehicle only 1.5T $\leq$ GVWR $\leq$ 3.5T with or without trailer	51 178	89.4	10.6
	Microcar (enclosed motorized quadricycle)	4 462	81.6	18.4

	Moped <50cc	23 481	61.7	38.3
	Motorcycle > 125cc	50 552	57.0	43.0
	Motorcycle > 50cc and ≤ 125cc	11 171	64.2	35.8
	Motorized personal transport device	11 674	83.0	17.0
	Non-motorized personal transport device	1 326	83.9	16.1
	Other vehicle	3 101	83.9	16.1
	Passenger car only	461 758	85.5	14.5
	Road tractor + semi-trailer	3 959	88.0	12.0
	Road tractor only	220	90.9	9.1
	Scooter < 50cc	18 967	82.0	18.0
	Scooter > 125cc	6 906	72.8	27.2
	Scooter > 50cc and ≤ 125cc	14 191	77.9	22.1
	Special vehicle	815	88.5	11.5
	Train	334	79.9	20.1
	Tramway	1 149	89.5	10.5
	Undeterminable	940	67.1	32.9
`obs`	Building wall bridge pier	9 922	56.3	43.7
	Concrete guardrail	11 505	81.3	18.7
	Culvert – culvert head	920	35.8	64.2
	Curb	6 000	69.6	30.4
	Ditch embankment rock wall	15 372	52.2	47.8
	Island refuge bollard	1 353	66.4	33.6
	Metal guardrail	11 110	69.4	30.6
	Not applicable	613 522	84.9	15.1
	Other fixed obstacle on roadway	5 412	76.9	23.1
	Other fixed obstacle on sidewalk or shoulder	4 829	62.5	37.5
	Other guardrail	1 486	64.6	35.4
	Parapet	942	58.8	41.2
	Parked vehicle	17 493	82.0	18.0
	Pole	9 811	64.3	35.7
	Run-off-road without obstacle	3 528	61.4	38.6
	Street furniture	3 965	77.9	22.1
	Tree	13 458	43.7	56.3
	Vertical road sign support or emergency call post	2 431	62.2	37.8
`obsm`	Domestic animal	603	68.5	31.5
	None	142 851	66.2	33.8
	Other	9 980	91.1	8.9
	Pedestrian	105 531	82.3	17.7
	Rail vehicle	709	73.8	26.2
	Vehicle	471 747	86.1	13.9
	Wild animal	1 638	66.9	33.1
`choc`	Front	267 395	77.0	23.0
	Front left	110 183	83.9	16.1
	Front right	94 956	84.4	15.6

	Left side	47 455	82.0	18.0
	Multiple impacts (rollovers)	12 138	63.8	36.2
	None	40 573	74.0	26.0
	Rear	72 983	92.4	7.6
	Rear left	25 270	93.0	7.0
	Rear right	20 350	92.2	7.8
	Right side	41 756	81.6	18.4
`manv`	Between two lanes	7 075	86.0	14.0
	Crossing the central reservation	1 287	63.1	36.9
	Crossing the roadway	17 443	81.1	18.9
	Door opening	2 078	98.8	1.2
	Driving on sidewalk	2 147	82.4	17.6
	Evasive maneuver	11 816	81.2	18.8
	In the bus lane opposite direction	287	84.3	15.7
	In the bus lane same direction	1 781	88.7	11.3
	Left (changing lane)	7 672	90.2	9.8
	Left (overtaking)	22 954	73.3	26.7
	Left (swerving)	31 819	63.7	36.3
	Left (turning)	56 849	91.5	8.5
	Making a U-turn on the roadway	4 509	92.3	7.7
	Merging	20 021	91.0	9.0
	No change of direction	313 473	79.4	20.6
	Other maneuvers	24 463	82.2	17.8
	Parked (with occupants)	2 861	91.3	8.7
	Parking maneuver	4 809	89.1	10.9
	Reversing	4 620	83.4	16.6
	Right (changing lane)	6 781	92.8	7.2
	Right (overtaking)	3 403	83.4	16.6
	Right (swerving)	15 967	70.0	30.0
	Right (turning)	19 552	89.0	11.0
	Same direction same lane	80 446	89.5	10.5
	Stopped (excluding parking)	18 056	94.5	5.5
	Unknown	42 858	74.8	25.2
	Wrong-way driving	8 032	69.5	30.5
`motor`	Electric	26 480	84.4	15.6
	Human-powered	31 947	73.2	26.8
	Hybrid electric	16 433	91.5	8.5
	Hydrocarbons	607 497	81.7	18.3
	Hydrogen	531	87.8	12.2
	Other	5 441	78.0	22.0
	Unknown	44 730	82.4	17.6
`senc`	Decreasing PK or PR or postal address number	249 377	81.1	18.9
	Increasing PK or PR or postal address number	330 530	81.9	18.1
	No reference point	109 124	82.7	17.3
	Unknown	44 028	81.3	18.7